Amartya Singh

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S-28/56-R-3A, Anula, Cantt, Varanasi
Github LinkedIn Portfolio

Education

Indraprastha Institute of Information Technology (IIITD) CGPA: 7.05
Bachelor of Technology (B.Tech), Computer Science and Social Sciences (CSSS)
2022 - 2026 (Expected)

Dhruva Public School, New Delhi Percentage: 92.20%
CBSE, Standard 12
2020 - 2022

Sant Atulanand Convent School, Varanasi Percentage: 95.00%
CBSE, Standard 10
2018 - 2020

Skills

Expertise Area: Software Engineering, Natural Language Processing (NLP), RAG Systems, Full-Stack Development, Systems Programming, Computer Architecture, Data Structures & Algorithms, Computer Vision, UI/UX Design, Deep Learning, Data Analysis.

Languages: Python, Java, C, C++, Kotlin, JavaScript, HTML/CSS, SQL (MySQL), Bash

Tools & Technologies: Next.js, React, FastAPI, PyTorch, TensorFlow, LangChain, Transformers, Pandas, NumPy, MERN Stack, Gem5, ns-3, Git, Docker, Linux, Firebase, Android Studio, Nginx, PM2, ChromaDB, VS Code, Figma.

Technical Electives: Natural Language Processing, Operating Systems, Computer Architecture, Algorithm Design, Machine Learning, Database Management Systems.

Work Experience

Software Engineering - AI Intern, Bynd AI (May, 25 - Jun, 25)

- Architected a complex, multi-faceted evaluation pipeline to validate the core AI system incorporating 10+ stages of Computer Vision and LLM analysis (GPT-4/Claude).
- Automated performance validation using a gold-standard dataset, improving testing quality by 70-80% and directly boosting the underlying RAG pipeline's effectiveness by 15-20%.

Web Development Intern, Vena Energy Pvt. Ltd (May, 24 - Jul, 24)

- Engineered a full-stack resource management platform with Role-Based Access Control (RBAC) to serve over 300+ stakeholders across 5-6 national project sites.
- Centralized the secure distribution of critical assets, including project credentials and technical guidebooks, to improve operational efficiency and data consistency.

Research Support (Space Debris), IIIT Delhi (Jul, 23 - Jan, 24)
Advisor: Mr. Arun Balaji Buduru

- Spearheaded the investigation of alternative architectural solutions for a project on orbital debris management using neural networks.
- Formulated and tested novel data collection strategies, influencing the high-level technical direction for the team's paper (arXiv:2307.09938).

Projects

AskAlma: Advanced RAG Chatbot for IIIT-D Website
Team Size – 2
(Jan 25 — Apr 25)
GitHub Repository

Technology Stack: Python, LangChain, FastAPI, Next.js, Transformers, ChromaDB

- Engineered a multi-modal data pipeline to scrape, clean, and augment unstructured data (Website, Excel, Docs) for a 2,000+ student user base.
- Deployed a novel, multi-level retrieval system that improved query response accuracy by over 50% against non-structured data baselines.

Office of Tech Transfer & Management - Platform Development
Team Size – 2
(Jan 25 — Apr 25)
Web Platform
GitHub Repository

Technology Stack: MERN Stack, Kotlin, XML, Python (RAG), Nginx, PM2

- Led the full-stack development of the OTMT platform, heavily optimizing the stack to operate on a single resource-constrained server (2GB RAM).
- Launched a comprehensive solution with 4 core features (Web-Portal, Android App, Admin Panel and Agentic Chatbot), load-tested to serve a target of 2,000+ users.

Deception Detection in Diplomacy Dialogues
Team Size – 1
(Jan 25 — Apr 25)
GitHub Repository

Technology Stack: Python, TensorFlow/Keras, Pandas, FastText, GCNs

- Built a novel hybrid deep learning model combining a Transformer encoder and a GCN to model complex game dynamics.
- Attained a state-of-the-art Macro F1 of 0.61 (from 0.48 baseline) and Lie F1 of 0.30 (from 0.14 baseline), more than doubling detection accuracy.

Advanced NoC Design & Simulation
Team Size – 2
(Aug 24 — Dec 24)
GitHub Repository

Technology Stack: Gem5, HeteroGarnet, Python, SystemC

- Fabricated 3+ novel Network-on-Chip (NoC) configurations from scratch using HeteroGarnet.
- Benchmarked system performance by simulating diverse traffic distribution patterns (e.g., spiral, random, uniform) at 2 Gbps.

Network Simulation & Analysis in ns-3
Team Size – 2
(Sep 24 — Dec 24)
GitHub Repository

Technology Stack: C++, ns-3, Python (analysis), Gnuplot

- Modeled a 10-node star network topology in ns-3, evaluating 4 key performance metrics (throughput, packet drop, latency and jitter) over a 60-second simulation to analyze performance.

Implementation of Core OS Primitives
Team Size – 2
(Oct 23— Dec 23)
GitHub Repository

Technology Stack: C, Linux System Calls (mmap, fork, exec), procfs, GDB

- Architected core OS resource managers, including a memory allocator that reduced memory fragmentation by over 20% on average and a testing framework for 3+ CPU scheduling policies.

Positions of Responsibility

- **Convenor, Techspectra (IIITD Tech Fest)** – Orchestrated a 3-day flagship tech fest for 10k+ national participants, managing 15+ tech events and two major hackathons.
- **Club Coordinator, Cyfuse** – Led IIIT Delhi's largest tech research club (50+ active members), fostering participation in multi-domain research projects.

Awards and Achievements

JEE - Advanced 2022: Qualified in top 2 percentile.
Online Courses: Completed Full-Stack Web-Development (Udemy) and Android-Development Basics (Google).

Declaration: The above information is accurate and true to the best of available knowledge.

Amartya Singh(Date: July 1, 2025)